

Battery & Resistor Circuits

Goal.

- Pin down what current and ΔV actually mean, and the four rules that follow just from charge and energy conservation.
- Turn Ohm's law into a tool: given a resistor and a ΔV across it, get I .
- Work a single resistor both with an ideal battery and with a real one (internal resistance r), to see what r costs you.
- Derive $R_S = R_1 + R_2$ for series and $R_P = (1/R_1 + 1/R_2)^{-1}$ for parallel from the rules in §1 — not as formulas to memorize, but as consequences of “same current” and “same ΔV .”

1. Definitions

Current: _____

units: _____ = _____

Change in voltage, ΔV : _____

Series: elements connected _____

Parallel: elements connected _____

2. Four rules

- Elements in series have _____
- Elements in parallel have _____
- Current is _____ at a junction.
- Around any closed loop, the ΔV 's _____

Check. The last two rules are each a conservation law in disguise. Which one is charge conservation, and which one is energy conservation?

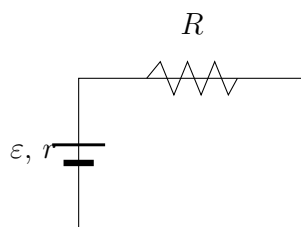
3. Ohm's law defines a resistor

$$\Delta V =$$

Given a resistor of resistance R and a voltage difference ΔV across it, the current through it is

$$I =$$

4. One resistor and a battery



(a) **Ideal battery:** $\varepsilon = 12 \text{ V}$, $r = 0$, $R = 4 \text{ } \Omega$.

$I =$

$V_B =$

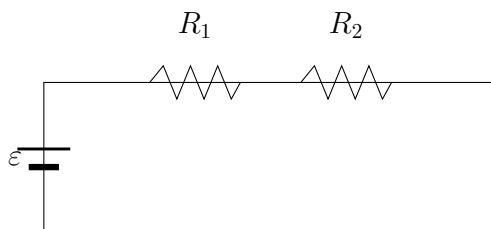
(b) **Real battery:** same ε and R , but now $r = 1 \text{ } \Omega$.

$I =$

$V_B =$

Check. V_B dropped in part (b). Where did that energy per charge go?

5. Two resistors in series



Derive R_S (symbols only — use the series current rule and Ohm's law on each resistor, then the loop rule):

start from $\varepsilon = \Delta V_1 + \Delta V_2$

$$R_S =$$

Now with numbers: $\varepsilon = 12 \text{ V}$, $R_1 = 4 \text{ } \Omega$, $R_2 = 8 \text{ } \Omega$ (ideal battery).

$$R_S =$$

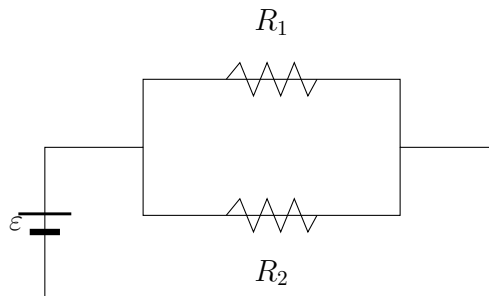
$$I =$$

$$\Delta V_1 =$$

$$\Delta V_2 =$$

Check. Check the loop rule with your two ΔV 's above: do they add up to ε ?

6. Two resistors in parallel



Derive R_P (symbols only — use the parallel voltage rule and Ohm's law on each resistor, then the junction rule at the top node):

start from $I = I_1 + I_2$

$$R_P =$$

Now with numbers: $\varepsilon = 12 \text{ V}$, $R_1 = 4 \text{ } \Omega$, $R_2 = 12 \text{ } \Omega$ (ideal battery).

$$R_P =$$

$$I_1 =$$

$$I_2 =$$

$$I_{\text{total}} =$$

Check. Which resistor carries more current, and why does that make sense given which one is smaller?